

EduBridge AI Platform Architecture

A Deep-Dive Technical Blueprint: Personalized Learning, Diagnostic Feedback, and Multimodal AI

Core Stack:	React 19, Node.js, Express, MongoDB Atlas, Google Gemini Flash	Status:	Active / Production-Ready
Target Domain:	K-12, Competitive Exams (JEE/NEET/UPSC), Higher Education	AI Engine:	Gemini 3.5 Flash Lite / 3.1 Flash / 3.6 Flash
Author / Lead:	Siddharth Tiwary (Solutions Architect)	Deployment:	Vercel (Frontend SPA) + Render (Backend REST API)

1. Executive Summary & System Vision

EduBridge AI is a state-of-the-art personalized diagnostic education platform engineered to eliminate generic, one-size-fits-all rote testing. Built on the modern **MERN Stack** (MongoDB, Express, React, Node.js) and tightly integrated with **Google Gemini Flash Multimodal AI**, EduBridge AI transitions testing from a passive grading mechanism into an active, actionable diagnostic feedback loop. The system pinpoints root causes of marks lost, calculates mathematical learning gap scores, generates adaptive micro-drills, and creates healthy 30-day study roadmaps.

The 5 Architectural Pillars of EduBridge AI:

- **Multimodal Marksheet OCR Analyzer:** Instant PDF/Image OCR extraction, grading trends, historical correlation.
- **Adaptive Diagnostic Test Engine:** Real-time single-call question generation with anti-duplication memory.
- **Precision Drill Test System:** Instant micro-quizzing targeted specifically at student diagnosed weak topics.
- **Automated Study & Wellness Planner:** Dynamic daily schedules with sleep, hydration, and revision milestones.
- **24/7 Contextual AI Mentor:** RAG-style interactive explanations grounded in student test and marksheet data.

2. Closed-Loop Adaptive Learning Architecture

EduBridge AI operates on a continuous feedback loop where every assessment directly updates the student's central learning profile:

Stage 1: Assessment & Ingestion	Student completes a Mock Test or uploads an existing semester/school Marksheet (PDF/Image).
Stage 2: Multimodal AI Diagnostics	Gemini Flash performs deep concept gap analysis, Bloom taxonomy grading, and mistake pattern classification.
Stage 3: Precision Drill Micro-Tests	Targeted 3-5 question adaptive drills with step-by-step reasoning isolate and resolve identified weak sub-topics.
Stage 4: Automated Study Roadmap	Generates personalized daily time blocks, healthy sleep/water schedules, and exports directly to iCalendar (.ics).
Stage 5: Progress Mastery Tracking	MongoDB Progress engine updates accuracy trendlines, strongest/weakest chapters, and exam readiness scores.

3. Multi-Tier System Architecture Blueprint

The platform is organized into 5 modular, decoupled architectural layers designed for sub-second execution, high security, and clean separation of concerns:

Layer	Component / Technologies	Responsibilities & Capabilities
1. Presentation Layer (Client SPA)	• React 19 & Vite 6 • Tailwind CSS v3 • Lucide React Icons • Recharts Data Visuals • Framer Motion • jsPDF & html2canvas	• Responsive Single Page Application (SPA) across desktop, tablet, and mobile. • Interactive diagnostic dashboards, marksheet OCR uploader, timed exam engine. • Client-side token storage, route protection, dynamic theme tokens, and PDF exports.
2. API Gateway & Security Layer	• Express Router Gateway • JWT Bearer Verification • Admin RBAC Middleware • Multer Disk Upload • CORS & URL Normalizer	• Centralized REST API routing mounted under /api/. • Stateless JWT authentication interceptor & admin privilege validation. • Secure file upload filtering (PDF, PNG, JPG up to 10MB) & global error handling.
3. Application & Controller Layer	• authController • testController • marksheetController • studyPlanController • progressController • adminController	• Handles user onboarding, session tokens, password reset flows. • Evaluates test scoring (+4/-1 competitive marking), speed rating, accuracy metrics. • Orchestrates OCR data persistence, target score calculation, and calendar sync. • Manages dynamic curriculum catalog (Subjects, Chapters, Topics, Questions).
4. AI Intelligence & Analytics Layer	• geminiService.js • marksheetService.js • gapScoreCalculator.js • duplicateChecker.js • Google Gemini Flash SDK	• Multi-model fallback pipeline (Gemini 3.5 Flash Lite / 3.1 Flash / 3.6 Flash). • Fast single-call question generation, prompt engineering, structured JSON parsing. • Multimodal vision OCR processing, learning gap scoring algorithm, mentor chat.
5. Data & Cloud Infrastructure	• MongoDB Atlas NoSQL • Mongoose ODM v8 • MongoMemoryServer • Vercel Edge Hosting • Render Cloud Web Service	• 12 normalized collections with indexing, pre-save hooks, and population links. • Automatic in-memory database fallback ensuring 100% test & startup reliability. • Vercel edge rewrite proxy for frontend SPA + Render managed backend API.

4. End-to-End Data Flow Pipeline



5. AI Intelligence Engine & Adaptive Drill Architecture

The core intelligence of EduBridge AI is powered by an ultra-fast, resilient **Google Gemini Flash** multi-model pipeline engineered for sub-2-second question generation and deep diagnostic performance evaluation.

A. High-Speed Gemini Multi-Model Fallback Hierarchy

Model Name	Target Latency	Primary Use Case	Resilience Priority
gemini-3.5-flash-lite	< 1,200 ms	Targeted Drill Tests, MCQ Generation, Fast Chat	Tier 1 (Primary)
gemini-3.1-flash-lite	< 1,300 ms	Study Roadmap Generation, Daily Schedule Logic	Tier 2 (Fallback)
gemini-3.6-flash	< 2,200 ms	Full Mock Exam Paper Setting, Bloom Analysis	Tier 3 (Fallback)
gemini-flash-latest	< 1,600 ms	Multimodal Marksheet OCR & Vision Extraction	Tier 4 (Fallback)

B. Single-Call Generation & Anti-Duplication Engine

Unlike naive implementations that make separate LLM calls per question, EduBridge AI utilizes an ultra-lean, structured JSON prompt that generates all 5–15 questions in a **single atomic request**. Before dispatching, the backend queries the student's recent question history and injects previously attempted questions into an anti-duplication exclusion prompt, ensuring 100% unique question sets across attempts.

C. Precision Adaptive Drill Test Workflow

Drill Trigger	Invoked from Marksheet Analysis or Mock Test results via the DrillTestModal component.
Context Extraction	Isolates exact weak topics (e.g. <i>Rotational Dynamics, Kinematics 2D</i>), historical accuracy, and missed concepts.
Dynamic Difficulty	Adaptive difficulty auto-adjusts based on previous drill accuracy: <50% → Easy, 50-80% → Medium, >80% → Hard.
Real-Time Feedback	Provides instant step-by-step conceptual reasoning, guiding hints, and formula breakdowns for every MCQ.
Mastery Sync	Submitting the drill updates the student's MongoDB Progress profile, transitioning topics from weak to strong.

D. Mathematical Learning Gap Score Formulation

EduBridge AI replaces arbitrary grading with a deterministic mathematical scoring formula computed in gapScoreCalculator.js:

Learning Gap Score Formula (0 to 100):
$$\text{GapScore} = (\text{AccuracyDeficit} \times 0.55) + (\text{ErrorRatio} \times 0.30) + (\text{SpeedPenalty} \times 0.15)$$

- Accuracy Deficit:** $100 - \text{Overall Accuracy (\%)}$
- Error Ratio:** $(\text{Wrong Count} / \text{Total Questions}) \times 100$
- Speed Penalty:** $\text{AvgTime} > 90\text{s} ? 25 : (\text{AvgTime} < 20\text{s} ? 15 : 0)$ (Penalizes rushing or stalling)
- Priority Thresholds:** GapScore ≥ 65: **Immediate Action** | 40–64: **High Priority** | <40: **Low Gap**

6. Marksheet OCR & Deep Academic Diagnostic Pipeline

The Marksheet Analyzer enables students to upload physical or digital grade cards, converting raw marks into personalized recovery plans, multi-semester comparisons, and calendar-synced study milestones.

Pipeline Stage	Technical Implementation	Output & Outcome
1. Ingestion & Validation	Multer disk storage in backend/uploads/marksheets. MIME type whitelist: PDF, PNG, JPG, JPEG (Max 10MB).	Validated local filepath and generated unique asset ID.
2. Multimodal OCR Extraction	Gemini Vision Multimodal API analyzes spatial layout, tables, subjects, marks obtained, max marks, and CGPA.	Structured JSON dataset of verified subjects, grades, and total percentages.
3. Historical Data Fusion	Backend merges current marksheet with student's MongoDB historical mock attempts, gap scores, and study plans.	Consolidated learning profile with cross-semester trend awareness.
4. Diagnostic AI Synthesis	Gemini Flash evaluates subject strengths, critical weaknesses, estimated study hours needed per chapter, and exam readiness.	Deep actionable insights report with root cause analysis.
5. 1-Click Recovery Plan	Generates structured 7-day, 14-day, and 30-day recovery plans targeting low-scoring subjects with specific revision blocks.	Direct sync to central StudyPlan module + downloadable .ics calendar file.
6. Target Score Projection	Mathematical simulator calculates exact marks required per subject to achieve a target percentage (e.g. 85%).	Actionable subject-by-subject required score trajectory.
7. Multi-Marksheet Comparison	Cross-compares 2+ uploaded marksheets across semesters/terms with delta metrics and trajectory graphs.	Visual improvement heatmaps and subject progression lines.

7. AI Mentor & Calendar Synchronization

24/7 Contextual AI Mentor Chat	Students can ask freeform questions about their marksheets or test mistakes (e.g. <i>'How do I bridge my 20-mark gap in Organic Chemistry?'</i>). The mentor service injects the student's exact attempt data, marksheet scores, and historical weakness logs into the system prompt for personalized, hyper-relevant coaching.
iCalendar (.ics) Export	Recovery plans generate RFC 5545 compliant .ics calendar feeds. Students can import their tailored revision blocks, mock test dates, and daily study slots directly into Google Calendar, Apple Calendar, or Outlook with one click.

8. Database Schema & Data Model Architecture

EduBridge AI uses **MongoDB Atlas** with **Mongoose ODM v8**. The schema is organized into 12 primary models with relational integrity:

Model / Collection	Key Fields & Schema Types	Relationships & Cardinality
User	name, email (unique), password (bcrypt), role ('student' 'admin'), examTarget, gradeClass, resetPasswordToken, resetPasswordExpire	1-to-Many with Attempt, MarksheetAnalysis, StudyPlan, Notification
Subject	name, category ('School' 'Engineering' 'Medical' 'Programming'), applicableExams[], applicableGrades[], description, icon	1-to-Many with Chapter, Topic, Question, MockTest
Chapter	subject (ObjectId), name, description	Belongs to Subject; 1-to-Many with Topic, Question
Topic	subject (ObjectId), chapter (ObjectId), name, description	Belongs to Subject and optional Chapter; 1-to-Many with Question
Question	user (ObjectId), subject, topic, chapter, questionText, options[4], correctAnswer, correctOptionIndex, explanation, difficulty, estimatedTimeSeconds, bloomLevel, questionType, conceptTested	Referenced by MockTest.questions[]; linked to User who generated/practiced it
MockTest	user (ObjectId), title, testType ('Topic' 'Chapter'), exam, grade, subject, topic, chapter, difficulty, questions[ObjectId], totalTimeSeconds, isCompleted	Belongs to User; contains Array of Question ObjectIds; 1-to-1 with Attempt
Attempt	user, mockTest, userAnswers[{question, selectedOptionIndex, isCorrect, timeSpentSeconds}], score, totalMarks, accuracy, correctCount, wrongCount, skippedCount, negativeMarks, percentile, speedRating, topicWiseAccuracy[], chapterWiseAccuracy[], difficultyWiseAccuracy[], learningGapScore, gapPriority	Belongs to User and MockTest; 1-to-1 with Feedback
Feedback	user, attempt (ObjectId), overallGrade, strengths[], weaknesses[], learningGaps[], topicWiseAnalysis[], chapterWiseAnalysis[], mistakeAnalysis, confidenceScore, estimatedStudyHours, examReadiness, improvementSuggestions[], studyStrategy, motivationalFeedback	Belongs to User; 1-to-1 with Attempt
Progress	user (ObjectId, unique), totalTestsTaken, totalQuestionsAttempted, totalCorrect, totalWrong, overallAccuracy, averageScore, currentStreak, lastTestDate, bestSubject, weakestSubject, strongestChapter, weakestChapter, weakTopics[{topicName, accuracy}], strongTopics[{topicName, accuracy}], overallLearningGapScore, examReadiness	1-to-1 per User; updated dynamically upon Mock/Drill test submissions
MarksheetAnalysis	user (ObjectId), fileName, fileUrl, fileType, fileSize, extractedData{studentName, rollNumber, schoolName, examName, subjects[], overallPercentage, cgpa}, aiAnalysis{summary, strengths[], weaknesses[], estimatedHoursPerChapter[]}, recoveryPlan, targetScoreAnalysis	Belongs to User; referenced during Study Plan auto-generation and mentor chat
StudyPlan	user (ObjectId, unique), examDate, availableHoursPerDay, targetScore, subject, topics[], weakAreas[], overview, dailyPlan[{timeSlot, activity, topicOrChapter, completed}], weeklyPlan[], revisionPlan[], mockTests[], healthyRoutine{sleepTime, wakeTime, exerciseTime, waterReminder, meditationTime, screenBreakInterval, healthyHabits[]}	1-to-1 active plan per User; synced with Mock Tests and Marksheet Recovery Plans
Notification	user (ObjectId), type ('Alert' 'Mastery' 'Weekly Report' 'Low Performance Alert'), title, message, isRead, createdAt	Belongs to User; generated automatically by diagnostic triggers

9. Security, API Routing Map & Deployment Infrastructure

EduBridge AI enforces a zero-trust stateless security model with robust error handling and production cloud hosting.

Router Prefix	HTTP Methods & Key Endpoints	Protection Level
/api/auth	POST /register, POST /login, GET /me, POST /forgot-password, POST /reset-password	Public / JWT (for /me)
/api/subjects	GET /, POST / (Admin), PUT /:id (Admin), DELETE /:id (Admin)	Public GET / Admin Only
/api/chapters	GET /?subjectId=x, POST / (Admin), PUT /:id (Admin), DELETE /:id (Admin)	Public GET / Admin Only
/api/topics	GET /?subjectId=x, POST / (Admin), PUT /:id (Admin), DELETE /:id (Admin)	Public GET / Admin Only
/api/tests	POST /generate, POST /drill/generate, POST /drill/submit, GET /:id, POST /submit, POST /mentor/chat	Protected (Student JWT)
/api/results	GET /user/history, GET /:attemptId	Protected (Student JWT)
/api/studyplan	GET /me, POST /generate, PUT /toggle-task	Protected (Student JWT)
/api/progress	GET /me	Protected (Student JWT)
/api/notifications	GET /, PUT /:id/read, PUT /read-all	Protected (Student JWT)
/api/marksheet	POST /upload-ocr, POST /save-analyze, GET /, GET /:id, DELETE /:id, POST /compare, POST /:id/target-score, POST /:id/sync-study-planner, POST /:id/chat, GET /:id/export-ics	Protected (Student JWT)
/api/admin	GET /users, PUT /users/:id, DELETE /users/:id, GET /questions, POST /questions, DELETE /questions/:id, GET /analytics	Admin Only (adminOnly)

10. Cloud Infrastructure & Verification Audit

Hosting Tier	Service & Configuration	High-Availability & Redundancy
Frontend	Vercel Edge Network (Global CDN)	Static build hosting with dynamic proxy rewrites to Render backend.
Backend API	Render Cloud Web Service (Node.js 24)	Auto-deploy from GitHub main branch with environment encryption.
Database	MongoDB Atlas Cloud Database Cluster	Automated MongoMemoryServer in-memory fallback for local dev/testing.
API Testing Status	32 / 32 Endpoints Passed (100%)	Automated integration test verified with zero route registration errors.